

Nirma University
Institute of Technology
Semester End Examination (IR), January - 2023
Master of Computer Applications, Semester - I
6CS102 Data Structures

Roll /
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.
5. Section wise separate answer book should be used.

SECTION-I

- Q-1. Answer the following. [16]**
- (A) **CO-4** Create Binary Search Tree (BST) by inserting the following list of (10)
BL-6 integers in order of their occurrence. Show all the steps.
65, 76, 63, 72, 53, 64, 87, 93, 80, 43, 58, 70, 74, 85
- (B) **CO-1** Given an array, **arr[1,2,...30][1,2,...10]** with base value **400** and (6)
BL-3 the size of each element is **4 Byte** in memory. Find the address
of **arr[28][5]** with the help of row-major order. Also find the
address of **arr[16][7]** with the help of column-major order.
- Q-2. Answer the following [18]**
- (A) **CO-2** Consider a circular queue of size six. Show the following (10)
BL-3 operations executed in sequence in circular queue by showing
the position of front and rear pointers at each step. Assume
queue is empty at initial state.
insert(A), insert(B), insert(C), insert(D), insert(E), delete(),
delete(), insert(F), insert(G), insert(H), insert(I), delete(), delete(),
delete().
- (B) **CO-2** Convert the following infix expression to postfix expression using (8)
BL-3 stack. Show all the intermediate steps.
 $8 + 4 / 2 * 3 - 6 + 7 - 8 ^ 1$
- Q-3. Answer the following [16]**
- (A) **CO-2** Write an algorithm of insertion and deletion of an elements in (10)
BL-3 circular link list. Insertion and deletion can be done at start, end
and at any position as mentioned by user.
- (B) **CO-2** Evaluate the following postfix expression using stack. Show all (6)
BL-5 the calculation steps.
 $8\ 7\ 6\ 3\ /\ 4\ 2\ /\ /\ +\ /\$

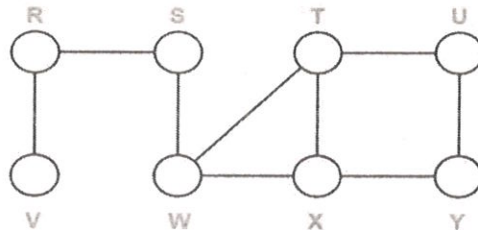
SECTION-II

Q-4. Answer the following. [16]

- (A) **CO-4 BL-6** Insert following elements into B+ tree of order 4. Construct the tree of right biased nature. Show the tree after every insertion. Also delete element 21 and 31 one by one from that created tree and show all the steps after deletion also. (10)
 1, 4, 7, 10, 17, 21, 31, 25, 19, 20, 28, 42
- (B) **CO-4 BL-6** Create the binary tree from the following traversal data. (6)
 In-order traversal: 6, 8, 9, 10, 12, 15, 17, 18, 20
 Post-order traversal: 20, 17, 18, 12, 15, 9, 6, 8, 10

Q-5. Answer the following [18]

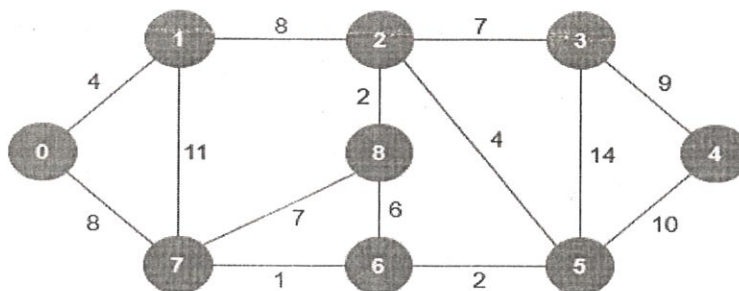
- (A) **CO-4 BL- 5** Consider the following Graph. Traverse the following graph using Breadth First Search (BFS) Technique and Depth First Search (DFS) technique. Show all the intermediate steps of traversals. Take starting vertex as "S" in both the cases. (10)



- (B) **CO-3 BL- 4** Apply the quick sort algorithm for sorting of the following data. Select pivot as the first element. Show the complete sorting step by step. (8)
 25, 19, 10, 30, 16, 35, 40, 39

Q-6. Answer the following [16]

- (A) **CO- 4 BL- 3** Apply Prim's algorithm for the following graph for finding minimum spanning tree. Show all the intermediate steps. Assume starting vertex as 7. (10)



- (B) **CO-3 BL- 3** Differentiate between linear and non linear data structure with a suitable example. Discuss with two real world application for the data structures identified by you. (6)